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# Math 32

## Trigonometry Fundamentals

# Math 32 Course at a Glance

## Trigonometry Fundamentals

### Trigonometric Foundations

| REQUIRED           |                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|
| <b>M32-FND-001</b> | I can use the Pythagorean theorem to determine an unknown side of a right triangle. |
| <b>M32-FND-002</b> | I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.              |
| <b>M32-FND-003</b> | I can convert angle measures between degrees and radians.                           |
| <b>M32-FND-004</b> | I can determine coterminal and reference angles for a given angle.                  |
| <b>M32-FND-005</b> | I can calculate arc length from a central angle and radius.                         |
| <b>M32-FND-006</b> | I can determine exact coordinates on the unit circle.                               |
| <b>M32-FND-007</b> | I can evaluate all six trigonometric functions exactly for common angles.           |
| <b>M32-FND-008</b> | I can solve right triangles using trigonometric ratios.                             |
| <b>M32-FND-009</b> | I can apply right-triangle trigonometry to contextual problems.                     |
| <b>M32-FND-010</b> | I can evaluate trigonometric functions from a point on a circle or terminal side.   |
| <b>M32-FND-011</b> | I can solve a basic trigonometric equation for all angles in a specified interval.  |

### Periodic Functions and Modeling

| REQUIRED           |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| <b>M32-GRF-001</b> | I can graph basic sine and cosine functions using key features.                       |
| <b>M32-GRF-002</b> | I can identify amplitude, midline, period, and phase shift from an equation or graph. |
| <b>M32-GRF-003</b> | I can graph transformed sine and cosine functions.                                    |
| <b>M32-GRF-004</b> | I can write sine and cosine equations that represent a periodic graph.                |
| <b>M32-GRF-005</b> | I can construct a sinusoidal model from contextual information or data.               |
| <b>M32-GRF-006</b> | I can interpret the parameters and outputs of a sinusoidal model in context.          |
| <b>M32-GRF-007</b> | I can graph tangent, secant, and cosecant functions with their asymptotes.            |

### Inverse Functions and Trigonometric Equations

| REQUIRED           |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| <b>M32-INV-001</b> | I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.         |
| <b>M32-INV-002</b> | I can explain how restricted domains make inverse trigonometric functions possible.   |
| <b>M32-INV-003</b> | I can solve trigonometric equations exactly or approximately on a specified interval. |
| <b>M32-INV-004</b> | I can express all solutions of a trigonometric equation in general form.              |
| <b>M32-INV-005</b> | I can evaluate compositions of trigonometric and inverse trigonometric functions.     |

### Non-Right Triangles

| REQUIRED           |                                                                              |
|--------------------|------------------------------------------------------------------------------|
| <b>M32-TRI-002</b> | I can solve a non-right triangle using the Law of Sines.                     |
| <b>M32-TRI-003</b> | I can determine the number of possible triangles in the ambiguous SSA case.  |
| <b>M32-TRI-004</b> | I can solve a non-right triangle using the Law of Cosines.                   |
| <b>M32-TRI-005</b> | I can select and apply an appropriate method to solve a triangle in context. |

# Math 32 Course at a Glance

## Trigonometry Fundamentals

### Trigonometric Identities

#### REQUIRED

|                    |                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M32-IDN-001</b> | I can rewrite trigonometric expressions using reciprocal and quotient identities, including $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$ . |
| <b>M32-IDN-002</b> | I can rewrite trigonometric expressions using the Pythagorean identities.                                                                        |
| <b>M32-IDN-003</b> | I can use even/odd and cofunction identities to rewrite trigonometric expressions.                                                               |
| <b>M32-IDN-004</b> | I can verify a trigonometric identity using the required identity families.                                                                      |

### Additional Trigonometric Identities

#### EXTENSION

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| <b>M32-IDX-001</b> | I can apply double-angle identities.                                   |
| <b>M32-IDX-002</b> | I can apply half-angle identities.                                     |
| <b>M32-IDX-003</b> | I can apply angle sum and difference identities.                       |
| <b>M32-IDX-004</b> | I can determine exact trigonometric values using extension identities. |

### Triangle Area from SAS

#### EXTENSION

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| <b>M32-TRX-001</b> | I can calculate the area of a triangle from two sides and the included angle. |
|--------------------|-------------------------------------------------------------------------------|

### Polar Coordinates

#### REQUIRED

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>M32-POL-001</b> | I can plot and identify points in polar coordinates.             |
| <b>M32-POL-002</b> | I can represent a polar point using equivalent coordinate pairs. |
| <b>M32-POL-003</b> | I can convert points between polar and rectangular coordinates.  |

# Math 32 Course at a Glance

## Trigonometry Fundamentals

| Polar Functions |                                                              |
|-----------------|--------------------------------------------------------------|
| EXTENSION       |                                                              |
| M32-PLX-001     | I can graph and interpret basic polar functions.             |
| M32-PLX-002     | I can convert equations between polar and rectangular forms. |

| Vectors     |                                                                  |
|-------------|------------------------------------------------------------------|
| EXTENSION   |                                                                  |
| M32-VEC-001 | I can represent a vector geometrically and in component form.    |
| M32-VEC-002 | I can add, subtract, and multiply vectors by scalars.            |
| M32-VEC-003 | I can determine vector magnitude, direction, and unit vector.    |
| M32-VEC-004 | I can resolve a vector into horizontal and vertical components.  |
| M32-VEC-005 | I can perform basic operations with vectors in three dimensions. |

| Parametric Equations |                                                                                 |
|----------------------|---------------------------------------------------------------------------------|
| EXTENSION            |                                                                                 |
| M32-PAR-001          | I can graph a plane curve defined by parametric equations with its orientation. |
| M32-PAR-002          | I can eliminate a parameter to obtain a rectangular equation.                   |
| M32-PAR-003          | I can create or interpret parametric equations for motion in the plane.         |

# Trigonometric Foundations

REQUIRED

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| M32-FND-001       | I can use the Pythagorean theorem to determine an unknown side of a right triangle.                            |
| SUPPORTING SKILLS |                                                                                                                |
| A                 | Extract perfect-power factors from a radical. (first introduced in Math 21)<br>SK-RAD-FACTOR-PERFECT · Apply   |
| R                 | Apply the Pythagorean theorem or its converse. (first introduced in Math 22)<br>SK-TRI-PYTHAGOREAN · Reinforce |

|                   |                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------|
| M32-FND-002       | I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.                                                         |
| SUPPORTING SKILLS |                                                                                                                                |
| A                 | Distinguish exact values from measured or rounded approximations. (first introduced in Math 22)<br>SK-GEO-EXACT-APPROX · Apply |
| A                 | Extract perfect-power factors from a radical. (first introduced in Math 21)<br>SK-RAD-FACTOR-PERFECT · Apply                   |
| R                 | Recall special-right-triangle ratios. (first introduced in Math 22)<br>SK-TRI-SPECIAL-RATIOS · Reinforce                       |

|                   |                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------|
| M32-FND-003       | I can convert angle measures between degrees and radians.                                                        |
| SUPPORTING SKILLS |                                                                                                                  |
| A                 | Add, subtract, multiply, and divide fractions. (inherited from middle school)<br>SK-NUM-FRACTION-OPERATE · Apply |
| I                 | Convert between degree and radian measure.<br>SK-TRIG-ANGLE-CONVERT · Introduce                                  |

|                   |                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------|
| M32-FND-004       | I can determine coterminal and reference angles for a given angle.                               |
| SUPPORTING SKILLS |                                                                                                  |
| I                 | Generate and recognize coterminal angles.<br>SK-TRIG-ANGLE-COTERMINAL · Introduce                |
| I                 | Determine a reference angle and its quadrant.<br>SK-TRIG-ANGLE-REFERENCE · Introduce             |
| I                 | Determine the sign of a trigonometric function by quadrant.<br>SK-TRIG-SIGN-QUADRANT · Introduce |

|                   |                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| M32-FND-005       | I can calculate arc length from a central angle and radius.                                                           |
| SUPPORTING SKILLS |                                                                                                                       |
| D                 | Calculate arc length from radius and central angle. (first introduced in Math 22)<br>SK-CIR-ARC-LENGTH · Deepen       |
| A                 | Select and report appropriate linear, square, or angular units. (first introduced in Math 22)<br>SK-GEO-UNITS · Apply |
| A                 | Convert between degree and radian measure. (first introduced in Math 32)<br>SK-TRIG-ANGLE-CONVERT · Apply             |

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| M32-FND-006       | I can determine exact coordinates on the unit circle.                                                          |
| SUPPORTING SKILLS |                                                                                                                |
| A                 | Recall special-right-triangle ratios. (first introduced in Math 22)<br>SK-TRI-SPECIAL-RATIOS · Apply           |
| A                 | Determine a reference angle and its quadrant. (first introduced in Math 32)<br>SK-TRIG-ANGLE-REFERENCE · Apply |
| I                 | Determine unit-circle coordinates for common angles.<br>SK-TRIG-UNIT-CIRCLE-COORD · Introduce                  |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

|                                                                                                             |                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <div>M32-FND-007</div> <div>I can evaluate all six trigonometric functions exactly for common angles.</div> |                                                                                                                            |
| SUPPORTING SKILLS                                                                                           |                                                                                                                            |
| I                                                                                                           | Express common trigonometric values exactly.<br>SK-TRIG-EXACT-VALUE · Introduce                                            |
| I                                                                                                           | Relate sine/cosecant, cosine/secant, and tangent/cotangent.<br>SK-TRIG-RECIPROCAL · Introduce                              |
| A                                                                                                           | Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32)<br>SK-TRIG-SIGN-QUADRANT · Apply |
| A                                                                                                           | Determine unit-circle coordinates for common angles. (first introduced in Math 32)<br>SK-TRIG-UNIT-CIRCLE-COORD · Apply    |

|                                                                                           |                                                                                                                                       |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <div>M32-FND-008</div> <div>I can solve right triangles using trigonometric ratios.</div> |                                                                                                                                       |
| SUPPORTING SKILLS                                                                         |                                                                                                                                       |
| D                                                                                         | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Deepen  |
| D                                                                                         | Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-ANGLE · Deepen |
| D                                                                                         | Use a trigonometric ratio to determine a missing side. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-SIDE · Deepen               |
| R                                                                                         | Label sides relative to a selected angle. (first introduced in Math 22)<br>SK-TRIG-TRIANGLE-LABEL · Reinforce                         |

|                                                                                                   |                                                                                                                                      |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <div>M32-FND-009</div> <div>I can apply right-triangle trigonometry to contextual problems.</div> |                                                                                                                                      |
| SUPPORTING SKILLS                                                                                 |                                                                                                                                      |
| A                                                                                                 | Restrict a model to meaningful contextual inputs. (first introduced in Math 12)<br>SK-MODEL-VALID-DOMAIN · Apply                     |
| A                                                                                                 | Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)<br>SK-REP-UNITS · Apply                     |
| D                                                                                                 | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Deepen |
| A                                                                                                 | Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-ANGLE · Apply |
| A                                                                                                 | Use a trigonometric ratio to determine a missing side. (first introduced in Math 22)<br>SK-TRI-TRIG-SOLVE-SIDE · Apply               |

|                                                                                                                     |                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <div>M32-FND-010</div> <div>I can evaluate trigonometric functions from a point on a circle or terminal side.</div> |                                                                                                                            |
| SUPPORTING SKILLS                                                                                                   |                                                                                                                            |
| A                                                                                                                   | Apply the Pythagorean theorem or its converse. (first introduced in Math 22)<br>SK-TRI-PYTHAGOREAN · Apply                 |
| I                                                                                                                   | Use coordinates and radius to evaluate trigonometric functions.<br>SK-TRIG-CIRCLE-POINT · Introduce                        |
| A                                                                                                                   | Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32)<br>SK-TRIG-RECIPROCAL · Apply    |
| A                                                                                                                   | Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32)<br>SK-TRIG-SIGN-QUADRANT · Apply |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

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Apply

|                   |                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| M32-FND-011       | I can solve a basic trigonometric equation for all angles in a specified interval.                                         |
| SUPPORTING SKILLS |                                                                                                                            |
| A                 | Translate between inequality and interval notation. (first introduced in Math 12)<br>SK-NOT-INTERVAL · Apply               |
| A                 | Determine a reference angle and its quadrant. (first introduced in Math 32)<br>SK-TRIG-ANGLE-REFERENCE · Apply             |
| I                 | Use reference angles and periodicity to find every solution in an interval.<br>SK-TRIG-EQUATION-INTERVAL · Introduce       |
| I                 | Isolate a trigonometric expression before solving.<br>SK-TRIG-EQUATION-ISOLATE · Introduce                                 |
| A                 | Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32)<br>SK-TRIG-SIGN-QUADRANT · Apply |

# Periodic Functions and Modeling

REQUIRED

M32-GRF-001

I can graph basic sine and cosine functions using key features.

SUPPORTING SKILLS

|   |                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------|
| D | Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21)<br>SK-REP-TABLE-GRAPH · Deepen  |
| I | Locate quarter-period key points of a periodic function.<br>SK-TRIG-GRAPH-KEYPOINTS · Introduce                         |
| A | Determine unit-circle coordinates for common angles. (first introduced in Math 32)<br>SK-TRIG-UNIT-CIRCLE-COORD · Apply |

M32-GRF-002

I can identify amplitude, midline, period, and phase shift from an equation or graph.

SUPPORTING SKILLS

|   |                                                                                         |
|---|-----------------------------------------------------------------------------------------|
| I | Determine amplitude and vertical reflection.<br>SK-TRIG-GRAPH-AMPLITUDE · Introduce     |
| I | Determine a sinusoid's midline and vertical shift.<br>SK-TRIG-GRAPH-MIDLINE · Introduce |
| I | Determine period from a graph or equation.<br>SK-TRIG-GRAPH-PERIOD · Introduce          |
| I | Determine horizontal shift from a graph or equation.<br>SK-TRIG-GRAPH-PHASE · Introduce |

M32-GRF-003

I can graph transformed sine and cosine functions.

SUPPORTING SKILLS

|   |                                                                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | Interpret transformations applied to function inputs. (first introduced in Math 31)<br>SK-FUNC-TRANSFORM-HORIZONTAL · Apply                             |
| D | Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31)<br>SK-FUNC-TRANSFORM-SCALE · Deepen |
| A | Interpret transformations applied to function outputs. (first introduced in Math 31)<br>SK-FUNC-TRANSFORM-VERTICAL · Apply                              |
| D | Locate quarter-period key points of a periodic function. (first introduced in Math 32)<br>SK-TRIG-GRAPH-KEYPOINTS · Deepen                              |

M32-GRF-004

I can write sine and cosine equations that represent a periodic graph.

SUPPORTING SKILLS

|   |                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------|
| D | Connect parameters in a formula to features of its graph. (first introduced in Math 21)<br>SK-REP-FORMULA-GRAPH · Deepen |
| D | Determine amplitude and vertical reflection. (first introduced in Math 32)<br>SK-TRIG-GRAPH-AMPLITUDE · Deepen           |
| D | Determine a sinusoid's midline and vertical shift. (first introduced in Math 32)<br>SK-TRIG-GRAPH-MIDLINE · Deepen       |
| D | Determine period from a graph or equation. (first introduced in Math 32)<br>SK-TRIG-GRAPH-PERIOD · Deepen                |
| D | Determine horizontal shift from a graph or equation. (first introduced in Math 32)<br>SK-TRIG-GRAPH-PHASE · Deepen       |



# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

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Introduce

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Reinforce

D

Deepen

A

Apply

|                   |                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| M32-GRF-005       | I can construct a sinusoidal model from contextual information or data.                                                            |
| SUPPORTING SKILLS |                                                                                                                                    |
| A                 | Restrict a model to meaningful contextual inputs. (first introduced in Math 12)<br>SK-MODEL-VALID-DOMAIN · Apply                   |
| A                 | Determine amplitude and vertical reflection. (first introduced in Math 32)<br>SK-TRIG-GRAPH-AMPLITUDE · Apply                      |
| A                 | Determine a sinusoid's midline and vertical shift. (first introduced in Math 32)<br>SK-TRIG-GRAPH-MIDLINE · Apply                  |
| A                 | Determine period from a graph or equation. (first introduced in Math 32)<br>SK-TRIG-GRAPH-PERIOD · Apply                           |
| I                 | Translate contextual maximum, minimum, cycle, and starting position into model parameters.<br>SK-TRIG-MODEL-PARAMETERS · Introduce |

|                   |                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M32-GRF-006       | I can interpret the parameters and outputs of a sinusoidal model in context.                                                                                  |
| SUPPORTING SKILLS |                                                                                                                                                               |
| A                 | Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)<br>SK-FUNC-INPUT-OUTPUT · Apply                        |
| D                 | Explain a model parameter in the language of its context. (first introduced in Math 21)<br>SK-MODEL-PARAMETER-INTERPRET · Deepen                              |
| A                 | Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)<br>SK-REP-UNITS · Apply                                              |
| D                 | Translate contextual maximum, minimum, cycle, and starting position into model parameters. (first introduced in Math 32)<br>SK-TRIG-MODEL-PARAMETERS · Deepen |

|                   |                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| M32-GRF-007       | I can graph tangent, secant, and cosecant functions with their asymptotes.                                                  |
| SUPPORTING SKILLS |                                                                                                                             |
| A                 | Interpret transformations applied to function inputs. (first introduced in Math 31)<br>SK-FUNC-TRANSFORM-HORIZONTAL · Apply |
| I                 | Locate vertical asymptotes of tangent, secant, and cosecant.<br>SK-TRIG-GRAPH-ASYMPTOTE · Introduce                         |
| A                 | Locate quarter-period key points of a periodic function. (first introduced in Math 32)<br>SK-TRIG-GRAPH-KEYPOINTS · Apply   |
| D                 | Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32)<br>SK-TRIG-RECIPROCAL · Deepen    |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

# Inverse Functions and Trigonometric Equations

REQUIRED

M32-INV-001

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

SUPPORTING SKILLS

A

Express common trigonometric values exactly. (first introduced in Math 32)  
SK-TRIG-EXACT-VALUE · Apply

I

Evaluate an inverse trigonometric expression exactly or approximately.  
SK-TRIG-INVERSE-EVALUATE · Introduce

I

Recall the principal-value ranges of inverse trigonometric functions.  
SK-TRIG-INVERSE-RANGE · Introduce

M32-INV-002

I can explain how restricted domains make inverse trigonometric functions possible.

SUPPORTING SKILLS

A

Exchange domain and range between a function and its inverse. (first introduced in Math 31)  
SK-FUNC-INVERSE-DOMAIN-RANGE · Apply

A

Apply the horizontal line test for invertibility. (first introduced in Math 31)  
SK-FUNC-INVERSE-HLT · Apply

A

Test whether a function is one-to-one on a domain. (first introduced in Math 31)  
SK-FUNC-ONE-TO-ONE · Apply

D

Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32)  
SK-TRIG-INVERSE-RANGE · Deepen

M32-INV-003

I can solve trigonometric equations exactly or approximately on a specified interval.

SUPPORTING SKILLS

A

Check a proposed solution in the original equation. (first introduced in Math 12)  
SK-ALG-CHECK-SOLUTION · Apply

A

Determine a reference angle and its quadrant. (first introduced in Math 32)  
SK-TRIG-ANGLE-REFERENCE · Apply

D

Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32)  
SK-TRIG-EQUATION-INTERVAL · Deepen

D

Isolate a trigonometric expression before solving. (first introduced in Math 32)  
SK-TRIG-EQUATION-ISOLATE · Deepen

A

Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32)  
SK-TRIG-INVERSE-EVALUATE · Apply

M32-INV-004

I can express all solutions of a trigonometric equation in general form.

SUPPORTING SKILLS

D

Generate and recognize coterminal angles. (first introduced in Math 32)  
SK-TRIG-ANGLE-COTERMINAL · Deepen

I

Express a periodic solution family in general form.  
SK-TRIG-EQUATION-GENERAL · Introduce

A

Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32)  
SK-TRIG-EQUATION-INTERVAL · Apply

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| M32-INV-005       | I can evaluate compositions of trigonometric and inverse trigonometric functions.                                                         |
| SUPPORTING SKILLS |                                                                                                                                           |
| A                 | Substitute one function formula into another. (first introduced in Math 31)<br>SK-FUNC-COMPOSE-FORMULA · Apply                            |
| A                 | Verify inverse functions using composition. (first introduced in Math 31)<br>SK-FUNC-INVERSE-VERIFY · Apply                               |
| D                 | Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32)<br>SK-TRIG-INVERSE-EVALUATE · Deepen |
| A                 | Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32)<br>SK-TRIG-INVERSE-RANGE · Apply      |

# Non-Right Triangles

REQUIRED

M32-TRI-002

I can solve a non-right triangle using the Law of Sines.

SUPPORTING SKILLS

|   |                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------|
| A | Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32)<br>SK-TRIG-INVERSE-EVALUATE · Apply |
| I | Apply the Law of Sines to determine missing sides or angles.<br>SK-TRIG-LAW-SINES · Introduce                                            |
| D | Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32)<br>SK-TRIG-TRIANGLE-DATA · Deepen                    |

M32-TRI-003

I can determine the number of possible triangles in the ambiguous SSA case.

SUPPORTING SKILLS

|   |                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------|
| I | Test and interpret the possible outcomes of SSA data.<br>SK-TRIG-AMBIGUOUS-CASE · Introduce                              |
| D | Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32)<br>SK-TRIG-LAW-SINES · Deepen |
| A | Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32)<br>SK-TRIG-TRIANGLE-DATA · Apply     |

M32-TRI-004

I can solve a non-right triangle using the Law of Cosines.

SUPPORTING SKILLS

|   |                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------|
| A | Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32)<br>SK-TRIG-INVERSE-EVALUATE · Apply |
| I | Apply the Law of Cosines to determine missing sides or angles.<br>SK-TRIG-LAW-COSINES · Introduce                                        |
| D | Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32)<br>SK-TRIG-TRIANGLE-DATA · Deepen                    |

M32-TRI-005

I can select and apply an appropriate method to solve a triangle in context.

SUPPORTING SKILLS

|   |                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------|
| A | Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)<br>SK-REP-UNITS · Apply                    |
| A | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Apply |
| A | Apply the Law of Cosines to determine missing sides or angles. (first introduced in Math 32)<br>SK-TRIG-LAW-COSINES · Apply         |
| A | Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32)<br>SK-TRIG-LAW-SINES · Apply             |
| D | Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32)<br>SK-TRIG-TRIANGLE-DATA · Deepen               |
| I | Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data.<br>SK-TRIG-TRIANGLE-METHOD · Introduce           |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

# Trigonometric Identities

REQUIRED

M32-IDN-001

I can rewrite trigonometric expressions using reciprocal and quotient identities, including  $\tan(\theta) = \sin(\theta)/\cos(\theta)$ .

SUPPORTING SKILLS

|   |                                                                                               |
|---|-----------------------------------------------------------------------------------------------|
| I | Apply reciprocal and quotient identities.<br>SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT · Introduce |
| I | Substitute an identity to rewrite an expression.<br>SK-TRIG-IDENTITY-SUBSTITUTE · Introduce   |

M32-IDN-002

I can rewrite trigonometric expressions using the Pythagorean identities.

SUPPORTING SKILLS

|   |                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------|
| I | Apply the Pythagorean trigonometric identities.<br>SK-TRIG-IDENTITY-PYTHAGOREAN · Introduce                            |
| D | Substitute an identity to rewrite an expression. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUBSTITUTE · Deepen |

M32-IDN-003

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

SUPPORTING SKILLS

|   |                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------|
| D | Test a formula or graph for even or odd symmetry. (first introduced in Math 31)<br>SK-FUNC-SYMMETRY · Deepen           |
| I | Apply sine/cosine cofunction identities.<br>SK-TRIG-IDENTITY-COFUNCTION · Introduce                                    |
| I | Apply even and odd trigonometric identities.<br>SK-TRIG-IDENTITY-EVEN-ODD · Introduce                                  |
| D | Substitute an identity to rewrite an expression. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUBSTITUTE · Deepen |

M32-IDN-004

I can verify a trigonometric identity using the required identity families.

SUPPORTING SKILLS

|   |                                                                                                                                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------|
| A | Simplify an algebraic expression using operation properties. (inherited from middle school)<br>SK-ALG-EXPRESSION-SIMPLIFY · Apply |
| I | Choose a productive identity or algebraic manipulation.<br>SK-TRIG-IDENTITY-STRATEGY · Introduce                                  |
| D | Substitute an identity to rewrite an expression. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUBSTITUTE · Deepen            |
| I | Maintain equivalence while transforming one side of an identity.<br>SK-TRIG-IDENTITY-VERIFY · Introduce                           |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

## Additional Trigonometric Identities

EXTENSION

M32-IDX-001

I can apply double-angle identities.

SUPPORTING SKILLS

|   |                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------|
| I | Apply double-angle identities.<br>SK-TRIG-IDENTITY-DOUBLE-ANGLE · Introduce                                           |
| A | Substitute an identity to rewrite an expression. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUBSTITUTE · Apply |

M32-IDX-002

I can apply half-angle identities.

SUPPORTING SKILLS

|   |                                                                                                       |
|---|-------------------------------------------------------------------------------------------------------|
| A | Apply double-angle identities. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply |
| I | Apply half-angle identities.<br>SK-TRIG-IDENTITY-HALF-ANGLE · Introduce                               |

M32-IDX-003

I can apply angle sum and difference identities.

SUPPORTING SKILLS

|   |                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------|
| A | Substitute an identity to rewrite an expression. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUBSTITUTE · Apply |
| I | Apply angle sum and difference identities.<br>SK-TRIG-IDENTITY-SUM-DIFFERENCE · Introduce                             |

M32-IDX-004

I can determine exact trigonometric values using extension identities.

SUPPORTING SKILLS

|   |                                                                                                                     |
|---|---------------------------------------------------------------------------------------------------------------------|
| D | Express common trigonometric values exactly. (first introduced in Math 32)<br>SK-TRIG-EXACT-VALUE · Deepen          |
| A | Apply double-angle identities. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply               |
| A | Apply half-angle identities. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-HALF-ANGLE · Apply                   |
| A | Apply angle sum and difference identities. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-SUM-DIFFERENCE · Apply |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

# Triangle Area from SAS

EXTENSION

|                   |                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------|
| M32-TRX-001       | I can calculate the area of a triangle from two sides and the included angle.                             |
| SUPPORTING SKILLS |                                                                                                           |
| A                 | Express common trigonometric values exactly. (first introduced in Math 32)<br>SK-TRIG-EXACT-VALUE · Apply |
| I                 | Calculate triangle area from two sides and the included angle.<br>SK-TRIG-TRIANGLE-AREA-SAS · Introduce   |
| I                 | Classify triangle data as SSS, SAS, ASA, AAS, or SSA.<br>SK-TRIG-TRIANGLE-DATA · Introduce                |

# Math 32 In-Depth Curriculum

## Trigonometry Fundamentals

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

# Polar Coordinates

REQUIRED

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| M32-POL-001       | I can plot and identify points in polar coordinates.                                                           |
| SUPPORTING SKILLS |                                                                                                                |
| I                 | Plot a point from radius and angle.<br>SK-POL-PLOT · Introduce                                                 |
| A                 | Determine a reference angle and its quadrant. (first introduced in Math 32)<br>SK-TRIG-ANGLE-REFERENCE · Apply |

|                   |                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| M32-POL-002       | I can represent a polar point using equivalent coordinate pairs.                                            |
| SUPPORTING SKILLS |                                                                                                             |
| I                 | Generate equivalent polar coordinate pairs.<br>SK-POL-EQUIVALENT · Introduce                                |
| A                 | Generate and recognize coterminal angles. (first introduced in Math 32)<br>SK-TRIG-ANGLE-COTERMINAL · Apply |

|                   |                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| M32-POL-003       | I can convert points between polar and rectangular coordinates.                                                                          |
| SUPPORTING SKILLS |                                                                                                                                          |
| I                 | Use coordinate relationships to convert a point between systems.<br>SK-POL-CONVERT-POINT · Introduce                                     |
| A                 | Use coordinates and radius to evaluate trigonometric functions. (first introduced in Math 32)<br>SK-TRIG-CIRCLE-POINT · Apply            |
| A                 | Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32)<br>SK-TRIG-INVERSE-EVALUATE · Apply |



# Polar Functions

EXTENSION

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| M32-PLX-001       | I can graph and interpret basic polar functions.                                                                                       |
| SUPPORTING SKILLS |                                                                                                                                        |
| A                 | Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)<br>SK-FUNC-INPUT-OUTPUT · Apply |
| D                 | Plot a point from radius and angle. (first introduced in Math 32)<br>SK-POL-PLOT · Deepen                                              |
| A                 | Locate quarter-period key points of a periodic function. (first introduced in Math 32)<br>SK-TRIG-GRAPH-KEYPOINTS · Apply              |

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| M32-PLX-002       | I can convert equations between polar and rectangular forms.                                                                      |
| SUPPORTING SKILLS |                                                                                                                                   |
| A                 | Simplify an algebraic expression using operation properties. (inherited from middle school)<br>SK-ALG-EXPRESSION-SIMPLIFY · Apply |
| I                 | Substitute coordinate identities to convert an equation between systems.<br>SK-POL-CONVERT-EQUATION · Introduce                   |
| A                 | Apply the Pythagorean trigonometric identities. (first introduced in Math 32)<br>SK-TRIG-IDENTITY-PYTHAGOREAN · Apply             |

# Vectors

## EXTENSION

M32-VEC-001

I can represent a vector geometrically and in component form.

SUPPORTING SKILLS

|   |                                                                                                                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------|
| A | Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22)<br>SK-GEO-DIAGRAM-MARK · Apply |
| I | Write a vector in component form.<br>SK-VEC-COMPONENTS · Introduce                                                                     |

M32-VEC-002

I can add, subtract, and multiply vectors by scalars.

SUPPORTING SKILLS

|   |                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------|
| A | Add, subtract, multiply, and divide signed integers. (inherited from middle school)<br>SK-NUM-INTEGGER-OPERATE · Apply |
| I | Perform component-wise vector operations.<br>SK-VEC-OPERATE · Introduce                                                |

M32-VEC-003

I can determine vector magnitude, direction, and unit vector.

SUPPORTING SKILLS

|   |                                                                                                                                                               |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22)<br>SK-COORD-DISTANCE-PYTHAGOREAN · Apply |
| I | Determine a vector's direction angle with correct quadrant.<br>SK-VEC-DIRECTION · Introduce                                                                   |
| I | Calculate vector magnitude using the distance formula.<br>SK-VEC-MAGNITUDE · Introduce                                                                        |
| I | Normalize a nonzero vector to obtain a unit vector.<br>SK-VEC-UNIT · Introduce                                                                                |

M32-VEC-004

I can resolve a vector into horizontal and vertical components.

SUPPORTING SKILLS

|   |                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------|
| A | Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22)<br>SK-TRI-TRIG-RATIO · Apply |
| D | Write a vector in component form. (first introduced in Math 32)<br>SK-VEC-COMPONENTS · Deepen                                       |
| A | Determine a vector's direction angle with correct quadrant. (first introduced in Math 32)<br>SK-VEC-DIRECTION · Apply               |

M32-VEC-005

I can perform basic operations with vectors in three dimensions.

SUPPORTING SKILLS

|   |                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------|
| I | Represent and operate on vectors with three components.<br>SK-VEC-3D · Introduce                                  |
| D | Calculate vector magnitude using the distance formula. (first introduced in Math 32)<br>SK-VEC-MAGNITUDE · Deepen |
| D | Perform component-wise vector operations. (first introduced in Math 32)<br>SK-VEC-OPERATE · Deepen                |

# Parametric Equations

EXTENSION

|                   |                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| M32-PAR-001       | I can graph a plane curve defined by parametric equations with its orientation.                                                        |
| SUPPORTING SKILLS |                                                                                                                                        |
| A                 | Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)<br>SK-FUNC-INPUT-OUTPUT · Apply |
| I                 | Graph a parametric curve with its direction of motion.<br>SK-PAR-GRAPH · Introduce                                                     |
| I                 | Generate ordered pairs from parametric equations.<br>SK-PAR-TABLE · Introduce                                                          |

|                   |                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| M32-PAR-002       | I can eliminate a parameter to obtain a rectangular equation.                                                            |
| SUPPORTING SKILLS |                                                                                                                          |
| A                 | Isolate a specified variable in an equation or formula. (first introduced in Math 12)<br>SK-ALG-VARIABLE-ISOLATE · Apply |
| I                 | Eliminate a parameter algebraically.<br>SK-PAR-ELIMINATE · Introduce                                                     |
| A                 | Substitute an equivalent expression for a variable. (first introduced in Math 12)<br>SK-SYS-SUBSTITUTE · Apply           |

|                   |                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| M32-PAR-003       | I can create or interpret parametric equations for motion in the plane.                                                         |
| SUPPORTING SKILLS |                                                                                                                                 |
| A                 | Explain a model parameter in the language of its context. (first introduced in Math 21)<br>SK-MODEL-PARAMETER-INTERPRET · Apply |
| I                 | Interpret or construct parametric equations for planar motion.<br>SK-PAR-MODEL · Introduce                                      |
| D                 | Generate ordered pairs from parametric equations. (first introduced in Math 32)<br>SK-PAR-TABLE · Deepen                        |
| A                 | Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12)<br>SK-REP-UNITS · Apply                |